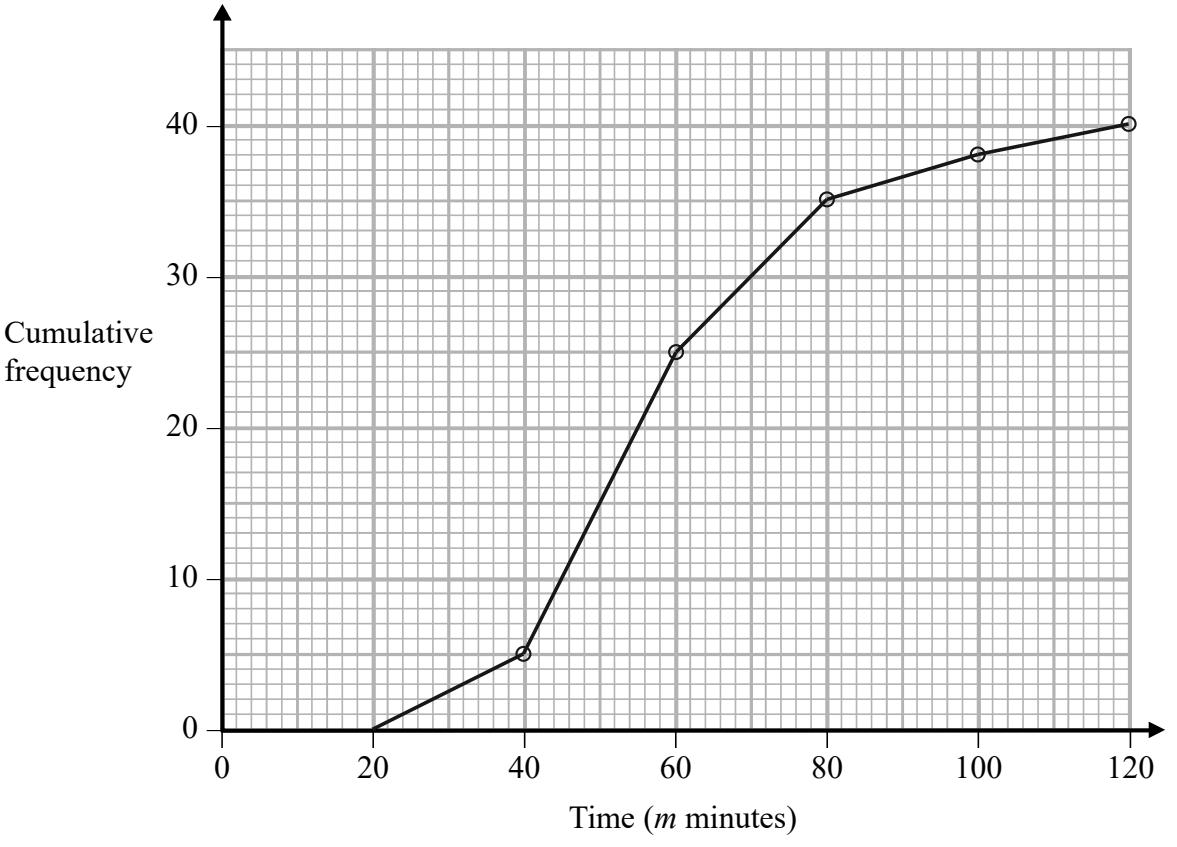
10 The cumulative frequency table shows information about the times, in minutes, taken by 40 people to complete a puzzle.

Time (m minutes)	Cumulative frequency
$20 < m \leqslant 40$	5
$20 < m \leqslant 60$	25
$20 < m \leqslant 80$	35
$20 < m \leqslant 100$	38
$20 < m \leqslant 120$	40

(a) On the grid below, draw a cumulative frequency graph for this information.



(2)

2 marks total
Plot upper-class points
Join in increasing order

(b) Use your graph to find an estimate for the interquartile range.

Lower quartile at CF = 10: about 45 min.

Upper quartile at CF = 30: about 70 min.

IQR ≈ 70 – 45 = 25 minutes.

2 marks total
Read quartiles at 10 and 30
IQR about 25 min

..... minutes

(2)

One of the 40 people is chosen at random.

(c) Use your graph to find an estimate for the probability that this person took between 50 minutes and 90 minutes to complete the puzzle.

At 50 min, CF ≈ 15; at 90 min, CF ≈ 36½.

Between: about 21½ of 40 people.

Probability ≈ 21½/40 = 0.54.

2 marks total
Subtract graph readings
Probability about 0.54

.....

(2)

(Total for Question 10 is 6 marks)